MATH 547: HOMEWORK 3
DUE ON: MONDAY, OCTOBER 21, 9AM.

Note: this is the longest homework of the semester. I strongly recommend that you start working on it early!

Problem 1, 10 points:

Let ξ be a Rademacher random variable, namely,

$$\mathbb{P}(\xi = 1) = \mathbb{P}(\xi = -1) = 1/2.$$

Show directly from the definition of sub-Gaussian random variables that $\xi \in \mathrm{SG}(1)$.

Definition of Sub-Gaussian variable X

$$\mathbb{E}\, e^{\lambda(X - \mathbb{E}X)} \le e^{\frac{\lambda^2\sigma^2}{2}}$$

When $\mathbb{E}X = 0$, we have $\mathbb{E}\, e^{\lambda X} \le e^{\frac{\lambda^2\sigma^2}{2}}$. $\mathbb{E}[\xi] = 0$

Solution: $\mathbb{E}\, e^{\lambda\xi}$ ($\xi \sim \mathrm{Ber}(1/2)$) $= P(\xi = 1)\, e^{\lambda} + P(\xi = -1)\, e^{-\lambda}$

$$= \frac{1}{2} e^{\lambda} + \frac{1}{2} e^{-\lambda} = \frac{1}{2}(e^{\lambda} + e^{-\lambda})$$

$$= \lim_{n\to\infty} \frac{1}{2}\left(\sum_{i=1}^{n} \frac{\lambda^i}{i!} + \sum_{i=1}^{n} \frac{(-\lambda)^i}{i!} \right)$$

$$= \frac{1}{2}\left(2\left\{ 1 + \frac{\lambda^2}{2!} + \frac{\lambda^4}{4!} + \cdots \right\} \right)$$

$$= \lim_{m\to\infty} \sum_{k=0}^{m} \frac{\lambda^{2m}}{(2m)!} = \lim_{m\to\infty} \sum_{k=0}^{m} \frac{(\lambda^2)^m}{(2m)!}$$

Now, $(2m)! = \underbrace{2m \cdot (2m-1) \cdot \cdots \cdot (m+1)}_{\ge 2^m} \cdot \underbrace{m \cdot (m-1) \cdot \cdots \cdot 1}_{m!}$

$$\ge 2^m m!$$

$$\Rightarrow \sum_{k=0}^{m} \frac{(\lambda^2)^m}{(2m)!} \le \sum_{k=0}^{m} \frac{(\lambda^2)^m}{2^m m!} = \sum_{k=0}^{m} \frac{(\lambda^2/2)^m}{m!}$$

$$\Rightarrow \lim_{m\to\infty} \frac{(\lambda^2)^m}{(2m)!} \le \lim_{m\to\infty} \sum_{k=0}^{m} \frac{(\lambda^2/2)^m}{m!} = \exp(\lambda^2/2)$$

$\Rightarrow \mathbb{E}\, e^{\lambda\xi} \le \exp\left(\frac{\lambda^2}{2}\right) \Rightarrow \xi$ is 1-Sub Gaussian.

Problem 2, 10 points:

Let $X_1, \ldots, X_n$ be i.i.d. normal random variables with mean 0 and variance σ^2. Prove that

$$\mathbb{E} \max_{j=1,\ldots,n} X_j \geq c\sigma\sqrt{\log(n)},$$

where $c > 0$ is some numerical constant (independent of n).

[Hint: the most straightforward (but not the only) way to proceed is to use the fact that, for a nonnegative random variable Z, $\mathbb{E}Z \geq \sup_{\delta>0} \delta P(Z \geq \delta)$].

Hint: $\mathbb{E}Z \geq \sup_{\delta>0} \delta P(Z \geq \delta)$.

This comes from Markov's inequality for $Z \geq 0$

$$P(Z \geq \delta) \leq \frac{\mathbb{E}Z}{\delta}$$

$$\Rightarrow \mathbb{E}Z \geq \delta P(Z \geq \delta) \quad \forall \text{ arbitrary } \delta > 0$$

$$\Rightarrow \mathbb{E}Z \geq \sup_{\delta} \delta P(Z \geq \delta)$$

Since $X_i \sim N(0, \sigma^2)$, $X_i \sim \sigma Y_i$ where $Y_i \sim N(0,1)$

We will show that $\mathbb{E} \max_j Y_j \geq c\sqrt{\log(n)}$.

After we show the above, we get :-

$X_i = \sigma Y_i \Rightarrow \max_i X_i = \sigma \max Y_i$

$(\sigma > 0)$

$$\Rightarrow \mathbb{E} \max_i X_i = \sigma \, \mathbb{E} \max_i Y_i$$

$$\Rightarrow \mathbb{E} \max_i X_i \geq \sigma \, c\sqrt{\log(n)}$$

which is the claimed inequality.

First, we notice that the inequality is manifestly true for $n = 1$ since $\max_{i=1} Y_i = Y_1$ & $\mathbb{E} \max_i Y_i = \mathbb{E} Y_1 = 0$

and $c\sqrt{\log n} = c\sqrt{\log 1} = 0$

$\therefore$ for $n = 1$ $\max_i Y_i = 0 \leq 0 = c\sqrt{\log n}$

Problem 3, 10 points:

Let $X_1, \ldots, X_n$ be (not necessarily independent) random variables such that

$$\max_{j=1,\ldots,n} (\mathbb{E}|X_j|^p)^{1/p} = \sigma_p < \infty$$

for some $p > 1$. Prove that

$$\mathbb{E} \max_{j=1,\ldots,n} |X_j| \leq n^{1/p} \sigma_p.$$

(hint: follow the similar argument that we used in class for sub-Gaussian random variables, but replace $t \mapsto e^{\lambda t}$ but another (which?) function)

$$\mathbb{E} \max_j |X_j| = \mathbb{E} \max_j \left(|X_j|^p\right)^{1/p}$$

$$= \mathbb{E} \left(\max_j |X_j|^p\right)^{1/p} \quad \because (\cdot)^{1/p} \text{ is increasing for } p > 0$$

$$\leq \left(\mathbb{E} \max_j |X_j|^p\right)^{1/p} \quad \because (\cdot)^{1/p} \text{ is concave + Jensen's inequality}$$

$$\leq \left(\mathbb{E} \sum_j |X_j|^p\right)^{1/p} \quad \because \forall i, a_i \geq 0 \Rightarrow \max_i a_i \leq \sum_i a_i$$

$$= \left(\sum_j \mathbb{E}|X_j|^p\right)^{1/p}$$

$$\leq \left(n \max_j \mathbb{E}|X_j|^p\right)^{1/p}$$

$$= n^{1/p} \left(\max_j \mathbb{E}|X_j|^p\right)^{1/p}$$

$$= n^{1/p} \max_j \left(\mathbb{E}|X_j|^p\right)^{1/p} \quad \because (\cdot)^{1/p} \text{ is increasing.}$$

$$= n^{1/p} \sigma_p$$

$$\Rightarrow \mathbb{E} \max_j |X_j| \leq n^{1/p} \sigma_p \quad \blacksquare$$

Problem 4, 25 points:

Let $X \in \mathrm{SG}(\sigma^2)$ be a sub-Gaussian random variable. Prove that

(a) (5pts) $\mathrm{Var}(X) \le \sigma^2$;

(b) (10pts) Show that for any integer $p \ge 1$, $\mathbb{E}|X|^p \le p\, 2^{p/2}\, \sigma^p\, \Gamma(p/2)$, and deduce that

$$\|X\|_p := (\mathbb{E}|X|^p)^{1/p} \le C\sigma\sqrt{p} \tag{1}$$

for an absolute constant C that does not depend on σ or p (give a specific value of C). Here, $\Gamma(x)$ is the gamma function defined as

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt.$$

(c) (10pts) Conversely, show that if X is centered (that is, $\mathbb{E}X = 0$) and satisfies (1) for all $p \ge 1$, then X is sub-Gaussian with sub-Gaussian parameter $C_1\sigma$ for some constant $C_1 > 0$.

$X \in SG(\sigma^2) \Leftrightarrow \mathbb{E}\, e^{\lambda(X-\mathbb{E}X)} \le e^{\frac{\lambda^2\sigma^2}{2}}$

Let $Y = X - \mathbb{E}X$. $\mathbb{E}Y = \mathbb{E}X - \mathbb{E}X = 0$

$\therefore\ \mathbb{E}\, e^{\lambda Y} \le e^{-\frac{\lambda^2\sigma^2}{2}}$

Also, $\mathrm{Var}(Y) = \mathrm{Var}(X - \mathbb{E}X) = \mathrm{Var}(X)$

Let $\Psi(\lambda) = \mathbb{E}\, e^{\lambda Y}$ { moment generating function of Y }

$\Psi(\lambda) \le e^{\frac{\lambda^2\sigma^2}{2}}$

$\Psi'(\lambda) = \frac{d}{d\lambda}\mathbb{E}\, e^{\lambda Y} = \mathbb{E}\frac{d}{d\lambda} e^{\lambda Y} = \mathbb{E}\, Y e^{\lambda Y}$

$\Psi'(0) = \mathbb{E}Y = 0$

Similarly $\Psi''(\lambda) = \mathbb{E}\, Y^2 e^{\lambda Y}$ & $\Psi''(0) = \mathbb{E}Y^2 = \mathrm{Var}(Y)$

$$\Psi(\lambda) = \Psi(0) + \Psi'(0)\lambda + \Psi''(0)\frac{\lambda^2}{2} + o(\lambda^2)$$

$$= \underbrace{\mathbb{E}\, e^{0\cdot Y}}_{1} + \underbrace{\mathbb{E}[Y]}_{0}\cdot\lambda + \mathrm{Var}(Y)\cdot\frac{\lambda^2}{2} + o(\lambda^2)$$

$$= 1 + 0 + \frac{\lambda^2}{2}\mathrm{Var}(Y) + o(\lambda^2)$$

$$e^{\lambda^2\sigma^2/2} = 1 + \frac{\lambda^2\sigma^2}{2} + o(\lambda^2)$$

Since, $\Psi(\lambda) \le e^{\lambda^2\sigma^2/2}$

we have $1 + \frac{\lambda^2}{2}\mathrm{Var}(Y) + o(\lambda^2) \le 1 + \frac{\lambda^2\sigma^2}{2} + o(\lambda^2)$

Since the above is true when $h \to 0$, we take limit on both sides.

$$1 + \frac{\lambda^2}{2}\mathrm{Var}(Y) + o(\lambda^2) \leq 1 + \frac{\lambda^2\sigma^2}{2} + o(\lambda^2)$$

$$\Rightarrow \frac{\lambda^2}{2}\mathrm{Var}(Y) \leq \frac{\lambda^2\sigma^2}{2}$$

$$\Rightarrow \mathrm{Var}(Y) \leq \sigma^2 \quad \Rightarrow \mathrm{Var}(X - \mathbb{E}X) \leq \sigma^2$$

$$\Rightarrow \mathrm{Var}(X) \leq \sigma^2$$

Alternatively:

$$\psi''(0) = \mathrm{Var}(Y) = \mathrm{Var}(X - \mathbb{E}X) = \mathrm{Var}(X)$$

$$\psi''(x) := \lim_{h\to 0} \frac{\psi(x+h) - 2\psi(x) + f(x-h)}{h^2}$$

(Symmetric standard second difference quotient)

$$\Rightarrow \psi''(0) = \lim_{h\to 0} \frac{\psi(h) - 2\overset{1}{\psi(0)} + \psi(-h)}{h^2}$$

$$\Rightarrow \mathrm{Var}(Y) = \lim_{h\to 0} \frac{\psi(h) - 2\cdot 1 + \psi(-h)}{h^2}$$

We know that $\psi(h) \leq e^{\frac{h^2\sigma^2}{2}}$ & also $\psi(-h) \leq e^{\frac{h^2\sigma^2}{2}}$

$$\therefore \mathrm{Var}(Y) \leq \lim_{h\to 0} \frac{2e^{\frac{h^2\sigma^2}{2}} - 2}{h^2}$$

$$= \lim_{h\to 0} \frac{2\left(e^{h^2\sigma^2/2} - 1\right)}{h^2}$$

$$= 2\lim_{h\to 0} \frac{1 + \frac{h^2\sigma^2}{2} + o(h^2) - 1}{h^2}$$

$$= 2\lim_{h\to 0} \frac{\sigma^2}{2} + 2\lim_{h\to 0} \frac{o(h^2)}{h^2} = \sigma^2$$

($\frac{o(h^2)}{h^2} \to 0$)

$$\Rightarrow \mathrm{Var}(Y) \leq \sigma^2 \Rightarrow \mathrm{Var}(X) \leq \sigma^2$$

b)

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(1) $$\|X\|_p := (\mathbb{E}|X|^p)^{1/p} \le C\sigma\sqrt{p}$$

for an absolute constant C that does not depend on σ or p (give a specific value of C). Here, $\Gamma(x)$ is the gamma function defined as

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt.$$

(c) (10pts) Conversely, show that if X is centered (that is, $\mathbb{E}X = 0$) and satisfies (1) for all $p \ge 1$, then X is sub-Gaussian with sub-Gaussian parameter $C_1\sigma$ for some constant $C_1 > 0$.

We use the following facts (to be proved later)

① $P(|X| \ge t) \le 2\exp\left(\frac{-t^2}{2\sigma^2}\right)$

② For any non-negative random variable Z

$$\mathbb{E}Z^p = \int_0^\infty p\, t^{p-1}\, P(Z \ge t)\, dt$$

③ $$\mathrm{II} = \int_0^\infty t^{p-1} \exp\left(-\frac{t^2}{2\sigma^2}\right) dt$$

$$= \sigma^p\, 2^{(p-2)/2} \int_0^\infty u^{(p-2)/2}\, e^{-u}\, du$$

$$= \sigma^p\, 2^{(p-2)/2}\, \Gamma(p/2)$$

where $\Gamma(\alpha) = \int_0^\infty u^{\alpha-1} e^{-u}\, du$

Using ① & ② :

$$\mathbb{E}|X|^p = \int_0^\infty p\, t^{p-1}\, \mathbb{P}(|X| \ge t)$$

$$\le \int_0^\infty p\, t^{p-1}\, 2\exp\left(-\frac{t^2}{2\sigma^2}\right) dt$$

$$= 2p \int_0^\infty t^{p-1} \exp\left(\frac{-t^2}{2\sigma^2}\right) dt$$

$$= 2p\, \sigma^p\, 2^{(p-2)/2}\, \Gamma(p/2) \quad (\text{from } ③)$$

Now, $\|X\|_p := \left(\mathbb{E}|X|^p\right)^{1/p} \le \left[2p\sigma^p 2^{p/2-1}\Gamma(p/2)\right]^{1/p}$

$$= \left[p\sigma^p 2^{p/2}\Gamma(p/2)\right]^{1/p}$$

$$= p^{1/p}\sigma 2^{1/2}\Gamma^{1/p}(p/2)$$

$$= \sqrt{2}\sigma p^{1/p}\Gamma^{1/p}(p/2)$$

The stirling's Inequality for gamma function :

$$C x^{x-1/2} e^{-x} \le \Gamma(x) \le C x^{x-1/2} e^{-x} e^{1/12x}$$

https://gubner.ece.wisc.edu/notes/GammaFunctionStirling.pdf

$$\therefore \Gamma(p/2) \le C (p/2)^{p/2-1/2} e^{-p/2} e^{1/6p}$$

$$\le C (p/2)^{p/2-1/2} e^{-1/2} e^{1/6} \quad (\because p \ge 1)$$

$$= \tilde{C} (p/2)^{p/2-1/2}$$

$$\Rightarrow \left\{\Gamma(p/2)\right\}^{1/p} = \left\{\tilde{C}(p/2)^{p/2-1/2}\right\}^{1/p}$$

$$= (\tilde{C})^{1/p} (p/2)^{1/2-1/2p}$$

$\tilde{C}$ is a constant. If $\tilde{C} \le 1$, then $(\tilde{C})^{1/p} \le 1$
If $\tilde{C} \ge 1$, $(\tilde{C})^{1/p} \le \tilde{C}$.

$$\therefore \tilde{C} \le \max\{1, \tilde{C}\} =: K$$

$$\therefore \left\{\Gamma(p/2)\right\}^{1/p} \le K (p/2)^{1/2-\frac{1}{2p}}$$

From online discussions, it seems $C = \sqrt{2\pi}$
& hence $K = \sqrt{2\pi}\, e^{-1/2} e^{1/6}$

$$\therefore \{\Gamma(p/2)\}^{1/p} \leq C\,(p/2)^{1/2 - 1/2p}$$

$$\therefore \|X\|_p \leq \sqrt{2}\,\sigma\, p^{1/p}\, \Gamma^{1/p}(p/2)$$

$$\leq \sqrt{2}\,\sigma\, p^{1/p}\, K\, (p/2)^{\frac{1}{2} - \frac{1}{2p}}$$

Let $f(x) = x^{1/x} \Rightarrow \log f(x) = \frac{1}{x}\log x$

$$\Rightarrow \frac{\partial}{\partial x}\log f(x) = \frac{1}{x^2} - \frac{1}{x^2}\log x = 0$$

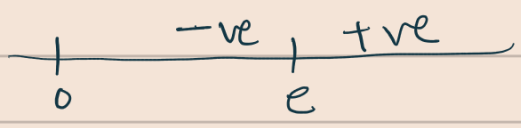
$$\Rightarrow \frac{1}{x^2} = \frac{1}{x^2}\log x$$

$$\Rightarrow \log x = 1$$

$$\Rightarrow x = e$$

$$\frac{\partial}{\partial x}\log f(x) = \frac{1}{x^2} - \frac{1}{x^2}\log x$$

$$= \frac{1}{x^2}(1 - \log x)$$



Hence $x = e$ is a maxima.

Hence $p^{1/p} < e^{1/e} \approx 1.44 < 2$

$$\therefore \|X\|_p < 2\sqrt{2}\,K\,\sigma\,\frac{\sqrt{p}}{\sqrt{2}}\,(p/2)^{-\frac{1}{2p}}$$

$$< 2K\,\sigma\sqrt{p}\;2^{1/2p}\cdot\frac{1}{p^{1/2p}}$$

For $p > 1$, $p^{1/2p} > 1 \Rightarrow \frac{1}{p^{1/2p}} < 1$

$$\therefore \|X\|_p < 2K\sigma\sqrt{p}\, 2^{\frac{1}{2p}} \cdot 1$$

$$\text{Also} \quad 2^{\frac{1}{2p}} = (\sqrt{2})^{\frac{1}{p}} \leq \sqrt{2} \quad \forall p \geq 1$$

$$\therefore \|X\|_p < \underbrace{2\sqrt{2}\,K}\sigma\sqrt{p}$$

$$\Rightarrow \|X\|_p < \tilde{K}\sigma\sqrt{p}$$

Now, we prove the three assumptions we made in the begining.

① $$\mathbb{P}(|X| \geq t) \leq 2\exp\left(\frac{-t^2}{2\sigma^2}\right)$$

This is a well known consequence of $X \in SG(\sigma^2)$

We will prove it here.

$$X \in SG(\sigma^2) \Rightarrow \mathbb{E}e^{\lambda X} \leq e^{\frac{\lambda^2\sigma^2}{2}}$$

$$P(X \geq t) = P(\underbrace{e^{\lambda X}}_{Y} \geq \underbrace{e^{\lambda t}}_{y})$$

By Markov inequality, for a non-negative random variable Y, $P(Y \geq y) \leq \frac{\mathbb{E}Y}{y}$

$$\therefore P(X \geq t) = P(e^{\lambda X} \geq e^{\lambda t}) = \frac{\mathbb{E}e^{\lambda X}}{e^{\lambda t}}$$

$$\Rightarrow P(X \geq t) = e^{-\lambda t}\mathbb{E}e^{\lambda X} \leq e^{\frac{\lambda^2\sigma^2}{2} - \lambda t}$$

$$\Rightarrow P(X \geq t) \leq e^{\frac{\lambda^2\sigma^2}{2} - \lambda t}$$

Now, this holds for all arbitrary λ. For a tighter bound, if $\tilde{\lambda} = \underset{\lambda}{\operatorname{argmin}}\, e^{\frac{\lambda^2\sigma^2}{2} - \lambda t}$, then

$$P(X \geq t) \leq \exp\left(\frac{\tilde{\lambda}^2\sigma^2}{2} - \tilde{\lambda} t\right)$$

$$0 = \frac{\partial}{\partial \lambda} \exp\left(\frac{\lambda^2\sigma^2}{2} - \lambda t\right) = \exp\left(\frac{\lambda^2\sigma^2}{2} - \lambda t\right)\left\{\lambda\sigma^2 - t\right\}$$

$$\Rightarrow \tilde{\lambda} = t/\sigma^2 \cdot$$ This is a minima since $\frac{\lambda^2\sigma^2}{2} - \lambda t$ is a upward parabola w.r.t λ.

$$\therefore P(X \geq t) \leq \exp\left(\frac{t^2\sigma^2}{\sigma^4 \cdot 2} - \frac{t}{\sigma^2} t\right)$$

$$= \exp\left(\frac{t^2}{2\sigma^2} - \frac{t^2}{\sigma^2}\right)$$

$$= \exp\left(-\frac{t^2}{2\sigma^2}\right)$$

$$\Rightarrow P(X \geq t) \leq \exp\left(\frac{-t^2}{2\sigma^2}\right)$$

$$X \in SG(\sigma^2) \iff -X \in SG(\sigma^2)$$

proof: $\mathbb{E}\, e^{\lambda X} \leq \exp(\lambda^2\sigma^2/2)$

$$\Rightarrow \mathbb{E}\, e^{-\lambda X} \leq \exp\left((-\lambda)^2\sigma^2/2\right) = \exp(\lambda^2\sigma^2/2)$$

$$\Rightarrow \mathbb{E}\, e^{\lambda Y} \le \exp(\lambda^2\sigma^2/2)$$

where $Y = -X$. $\therefore -X = Y \in SG(\sigma^2)$

Hence $P(X \le -t) = P(-X \ge t)$

$$= P(e^{-\lambda X} \ge e^{\lambda t})$$

$$\le \frac{\mathbb{E}\, e^{-\lambda X}}{e^{\lambda t}}$$

$$= \frac{e^{(-\lambda)^2\sigma^2/2}}{e^{\lambda t}}$$

$$= \exp\left(\frac{\lambda^2\sigma^2}{2} - \lambda t\right)$$

Similarly, $\tilde{\lambda} := \underset{\lambda}{\operatorname{argmin}} \exp\left(\frac{\lambda^2\sigma^2}{2} - \lambda t\right)$

$$\Rightarrow \tilde{\lambda} = t/\sigma^2$$

$$\therefore P(X \le -t) \le \exp\left(\frac{t^2\sigma^2}{\sigma^4 \cdot 2} - \frac{t^2}{\sigma^2}\right)$$

$$= \exp\left(\frac{t^2}{2\sigma^2} - \frac{t^2}{\sigma^2}\right)$$

$$= \exp\left(-\frac{t^2}{2\sigma^2}\right)$$

$$\therefore P(|X| \ge t) = P(\{X \ge t\} \cup \{X \le -t\})$$

$$\le P(X \ge t) + P(X \le -t)$$

Union bound

$$\therefore \mathbb{P}(|X| \geq t) \leq \exp\left(\frac{-t^2}{2\sigma^2}\right) + \exp\left(\frac{-t^2}{2\sigma^2}\right)$$

$$\mathbb{P}(|X| \geq t) \leq 2\exp\left(\frac{-t^2}{2\sigma^2}\right) \quad \forall t \geq 0$$

(2) If $Z \geq 0$, we show $\mathbb{E}Z^p = \int_0^\infty p t^{p-1} \mathbb{P}(Z \geq t)\, dt$

$$Z^p = \int_0^{Z^p} 1 \cdot du = \int_0^\infty \mathbb{1}\{u \leq Z^p\}\, du$$

$$= \int_0^\infty \mathbb{1}\{Z \geq u^{\frac{1}{p}}\}\, du$$

Change of variables $u = t^p$, $du = p t^{p-1} dt$

$$\therefore Z^p = \int_0^\infty \mathbb{1}\{Z \geq t\}\, p t^{p-1} dt$$

$$\mathbb{E}Z^p = \mathbb{E}\int_0^\infty \mathbb{1}\{Z \geq t\}\, p t^{p-1} dt$$

$$= \int_0^\infty \underline{\mathbb{E}\,\mathbb{1}\{Z \geq t\}}\, p t^{p-1} dt$$

Swapped integrals using Fubini-Tonelli.

$$= \int_0^\infty \overbrace{\mathbb{P}(Z \geq t)}\, p t^{p-1} dt$$

$$\therefore \mathbb{E}|X|^p = \int p t^{p-1} \mathbb{P}(|X| \geq t)\, dt$$

③ We will show $\mathrm{II} = \int_0^\infty t^{p-1} \exp\left(-\frac{t^2}{2\sigma^2}\right) dt$

$$= \sigma^p 2^{(p-2)/2} \underbrace{\int_0^\infty u^{(p-2)/2} e^{-u} du}_{\Gamma(p/2)}$$

$$\mathrm{II} = \int_0^\infty t^{p-1} \exp\left(\frac{-t^2}{2\sigma^2}\right) dt$$

Let $u = \frac{t^2}{2\sigma^2} \Rightarrow t = \sigma\sqrt{2u}$

$$\Rightarrow dt = \sqrt{2}\,\sigma \frac{1}{2\sqrt{u}} du = \frac{\sigma}{\sqrt{2u}} du$$

$$t^{p-1} = \sigma^{p-1} (2u)^{(p-1)/2}$$

$$\therefore \mathrm{II} = \int_0^\infty \sigma^{p-1} (2u)^{(p-1)/2} \exp(-u) \frac{\sigma}{\sqrt{2u}} du$$

$$= \int_0^\infty \sigma^{p-1} (2u)^{p/2} \frac{1}{\sqrt{2u}} \exp(-u) \frac{\sigma}{\sqrt{2u}} du$$

$$= \int_0^\infty \sigma^{p-1} (2u)^{p/2} \exp(-u) \frac{\sigma}{2u} du$$

$$= \int_0^\infty \sigma^p (2u)^{p/2-1} \exp(-u) du$$

$$= \sigma^p (2)^{p/2-1} \int_0^\infty u^{p/2-1} \exp(-u) du$$

Recall that $\Gamma(\alpha) = \int_0^\infty u^{\alpha-1} e^{-u} du$

$$\therefore \text{II} = \sigma^p (2)^{p/2-1} \Gamma(p/2)$$

$$= \sigma^p 2^{(p-2)/2} \Gamma(p/2)$$

$$\text{II} = \int_0^\infty t^{p-1} \exp\left(\frac{-t^2}{2\sigma^2}\right) dt$$

$$= \sigma^p 2^{(p-2)/p} \Gamma(p/2)$$

Problem 4, 25 points:

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$$\|X\|_p := (\mathbb{E}|X|^p)^{1/p} \le C\sigma\sqrt{p} \tag{1}$$

for an absolute constant C that does not depend on σ or p (give a specific value of C). Here, $\Gamma(x)$ is the gamma function defined as

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(c) (10pts) Conversely, show that if X is centered (that is, $\mathbb{E}X = 0$) and satisfies (1) for all $p \ge 1$, then X is sub-Gaussian with sub-Gaussian parameter $C_1\sigma$ for some constant $C_1 > 0$.

(c) $\mathbb{E}X = 0$ & $\|X\|_p := (\mathbb{E}|X|^p)^{1/p} \le C\sigma\sqrt{p}$

We want to show $\exists K$ such that

$$P(|X| \ge t) \le 2\exp\left(\frac{-t^2}{2K^2\sigma^2}\right) \quad \forall t > 0$$

We already showed that this is a consequence of $X \in SG(\sigma^2)$ but this is infact an equivalence i.e. the above implies $X \in SG(\sigma^2)$.

Since $|X| \geq 0$, by Markov's Inequality

$$P(|X| \geq t) = P(|X|^p \geq t^p) \leq \frac{\mathbb{E}|X|^p}{t^p}$$

Recall the premise $\|X\|_p = (\mathbb{E}|X|^p)^{1/p} \leq C\sigma\sqrt{p}$

$$\Rightarrow \mathbb{E}|X|^p \leq (C\sigma\sqrt{p})^p$$

$$\therefore P(|X| \geq t) \leq \left(\frac{C\sigma\sqrt{p}}{t}\right)^p$$

Let $\alpha = \frac{t}{C\sigma} \geq 0$

Then $P(|X| \geq t) \leq \left(\frac{\sqrt{p}}{\alpha}\right)^p \quad \forall\, p \geq 1$

Since the above works for any p, we want to minimize the RHS to get the tightest upper bound.

Let $g(p) = \log\left(\frac{\sqrt{p}}{\alpha}\right)^p = p\,\frac{1}{2}\log p - p\log\alpha$

$$0 = g'(p) = \frac{1}{2} + \frac{1}{2}\log p - \log\alpha$$

$$\Rightarrow \log p = (\log\alpha - 1/2)\cdot 2 = 2\log\alpha - 1 = \log\alpha^2 - 1$$

$$\Rightarrow p = \alpha^2/e$$

Let us pick $p = \frac{\alpha^2}{4e} \Rightarrow \sqrt{p} = \frac{\alpha}{2\sqrt{e}} \Rightarrow \frac{\sqrt{p}}{\alpha} = \frac{1}{2\sqrt{e}}$

$$\therefore P(|X| \geq t) \leq \left(\frac{1}{2\sqrt{e}}\right)^p$$

$$\Rightarrow \log P(|X| \geq t) \leq -p \log 2\sqrt{e} = -\frac{\alpha^2}{4e}\log 2\sqrt{e}$$

(p = $\alpha^2/4e$, $\log 2\sqrt{e}$ = K)

$\Rightarrow \log P(|X| \geq t) \leq -m\alpha^2$ for $m = \frac{1}{4e} \log 2\sqrt{e}$

$\Rightarrow P(|X| \geq t) \leq e^{-m\alpha^2} = e^{-m\frac{t^2}{c^2\sigma^2}}$

$\Rightarrow P(|X| \geq t) \leq e^{-\frac{t^2}{2K^2\sigma^2}}$ where $2K^2 = c^2/m$.

Hence $X \in SG(\sigma^2)$.